

One of the major concerns today is the pollution caused by the massive amounts of carbon dioxide, produced by modern means of transport by industry and so on. However, it's now possible to collect or capture some of this CO₂. Up to now, this captured carbon has mainly been stored beneath the ground. But although this is an effective way of dealing with it, the process is expensive. So increasingly, scientists are considering how this captured carbon can be put to some productive use or other. By recycling, this is already beginning to happen.

Let me give you some examples. Take the Netherlands country where the cultivation of flowers is big business. Growers there have already begun to use captured carbon in their greenhouses with great success. The carbon enriched air speeds growth by up to 30 %. It seems inevitable that the farmers who grow tomatoes, another important Dutch export will also soon be using captured carbon in their hot houses, too. I can give you examples from the United States, too. There captured carbon is already being successfully used by one firm to manufacture plastics. The carbon that is recycled in this way has a valuable job in that, it replaces other chemicals that would otherwise have been needed for the manufacturing process.

Another rather different American business has been making successful use of captured carbon in its production of medicines. This involves some fairly complicated processes, but the company is pleased with what it has done so far and confident that it will go on to achieve still more in the near future. More industries are poised to find their own ways of using captured carbon in the US as well. Progress in the use of captured carbon has also been made in China where power plants are being encouraged to capture their waste carbon dioxide. Carbon they have captured there has then been put to good use in metalworking processes, as well as in factories producing soft drinks, the income that such recycling generates is helping to pay for a further program of capturing and reusing carbon.

Let me now tell you about the cement industry. It is tackling the problem in a rather different way at present. This industry accounts for roughly 5 % of all human CO₂ emissions. This constitutes a larger proportion, even than that produced by the airline business, which is often seen by the public as the source of most of the CO₂ problems in the modern world. Why does the cement industry produce so much CO₂? Basically, it's because carbon is produced as a byproduct from the limestone used to make the cement. Moreover, the temperature required for the production of the cement has to be extremely high, resulting in even more CO₂ emissions.

Various companies all over the world are experimenting with alternative methods of cement production. In California, one firm is bubbling CO₂ from power plant chimneys through ash, which has been mixed with sea water. This produces a mineral which can be processed into cement. Here, CO₂ is used up instead of being produced.

Another company in London is investigating the use of alternative minerals to make cement, hoping to find one that can be roasted without releasing CO₂ into the atmosphere. It's had some success, one called serpentine. However, researchers are still investigating whether this type of cement will have enough strength for commercial purposes. These are but the beginnings. What of the future scientists are beginning to talk in very far reaching terms about the huge possibilities for captured carbon. Some say that it has the potential to provide enough power for entire cities. There's a whole range of ideas around. The main thrust of these is that it's time we stopped thinking of CO₂ as pollutant, and began considering it as a valuable resource. Among the many hopes for the future. A key one has to be that we may, at some point, be able to turn captured carbon back into fuel. Then we should indeed be able to feel that we are beginning to solve the problem of CO₂ emissions. Let me now turn to some other.